Bitcoin Mining & The Solar Opportunity Whitepaper Preview

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Whitepaper: Bitcoin Mining & The Solar Opportunity

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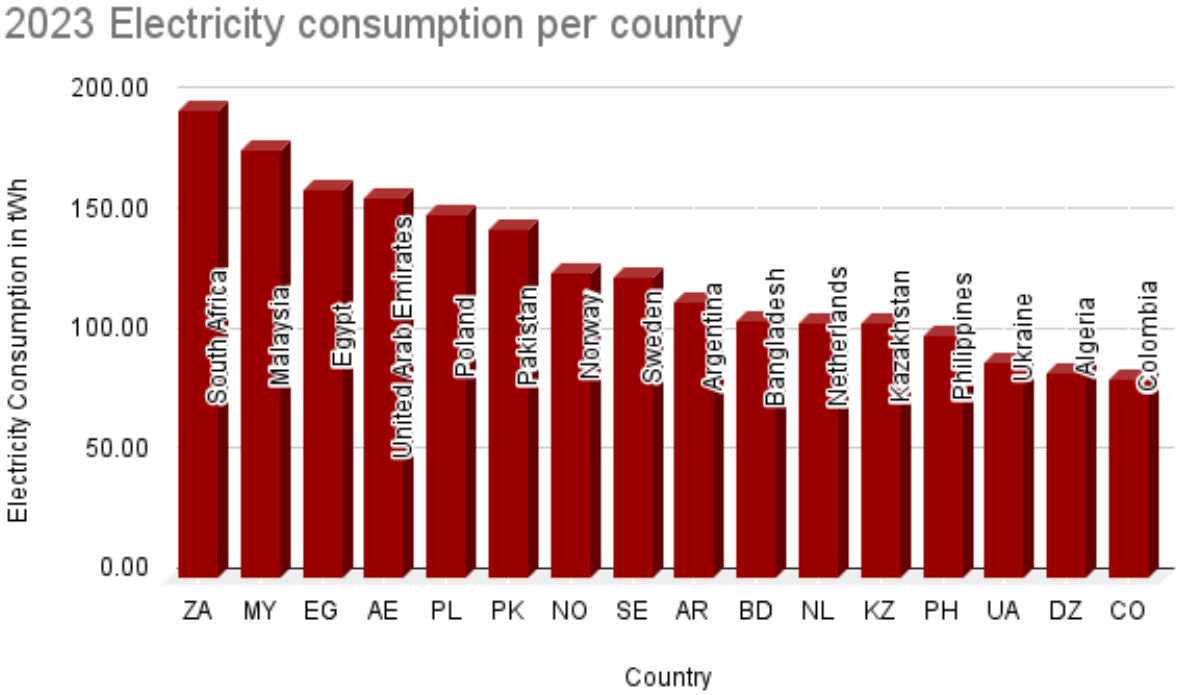
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Executive Summary

Seventeen years after Satoshi Nakamoto published the Bitcoin whitepaper, Bitcoin has evolved from a peer-to-peer electronic cash system into a globally distributed settlement network secured by proof-of-work computation. Public discourse around Bitcoin focuses on price volatility, speculation and often comparisons to gold and the S&P500, which have market caps of: **$1.8T**, **$31T** and **$58T** respectively (as of 2025). However, there is a more pertinent factor that needs addressing, and that is energy.

Bitcoin is valuable not because of speculation but because it is backed by a real-world annual estimated **120-200 terrawatt hours of high energy demand** required to secure the Bitcoin network. However, the same energy intensity that secures the network has become the primary constraint on mining profitability, regulatory acceptance, and long-term sustainability. Per the Cambridge Bitcoin Electricity Consumption Index, Bitcoin mining currently consumes an estimated **195.9 tWh**(Cambridge Bitcoin Electricity Consumption Index) of electricity annually(2025 range was 120-200 tWh)-more than Nordic countries like Norway(**124 tWh**)(World Population Review, 2026) as measured by the World Population Review's Electricity Consumption per country's 2023 data set.



Electricity Consumption per Country, 2023. Source: World Population Review (2023). *Visualization modified by author*

The process, which relies on non-renewables like coal for ~48% of global activity- emits an approximate 39M metric tonnes(Cambridge Digital Mining Industry Report, 2025) -98M metric tonnes(Swissinfo, 2025) of carbon dioxide annually, a greenhouse gas attributed to global warming. Furthermore, electricity-backed by energy, as an operational expense for bitcoin miners accounts for 60-80% of operational expenses(Solartechonline, 2025).

This whitepaper argues that the future competitiveness of Bitcoin mining depends on a structural shift toward low-cost, off-grid renewable energy sources in high-irradiance regions. Specifically, it evaluates solar energy deployment in Botswana as a strategic opportunity for Bitcoin miners seeking to reduce operating costs, mitigate regulatory risk, and improve long-term economic resilience.

This paper:

- Quantifies Bitcoin mining's energy requirements and cost structure
- Evaluates the economic case for solar-powered, off-grid mining
- Models a Botswana-based solar microgrid achieving ~**$0.08/kWh LCOE**
- Proposes a scalable roadmap for solar-powered compute infrastructure

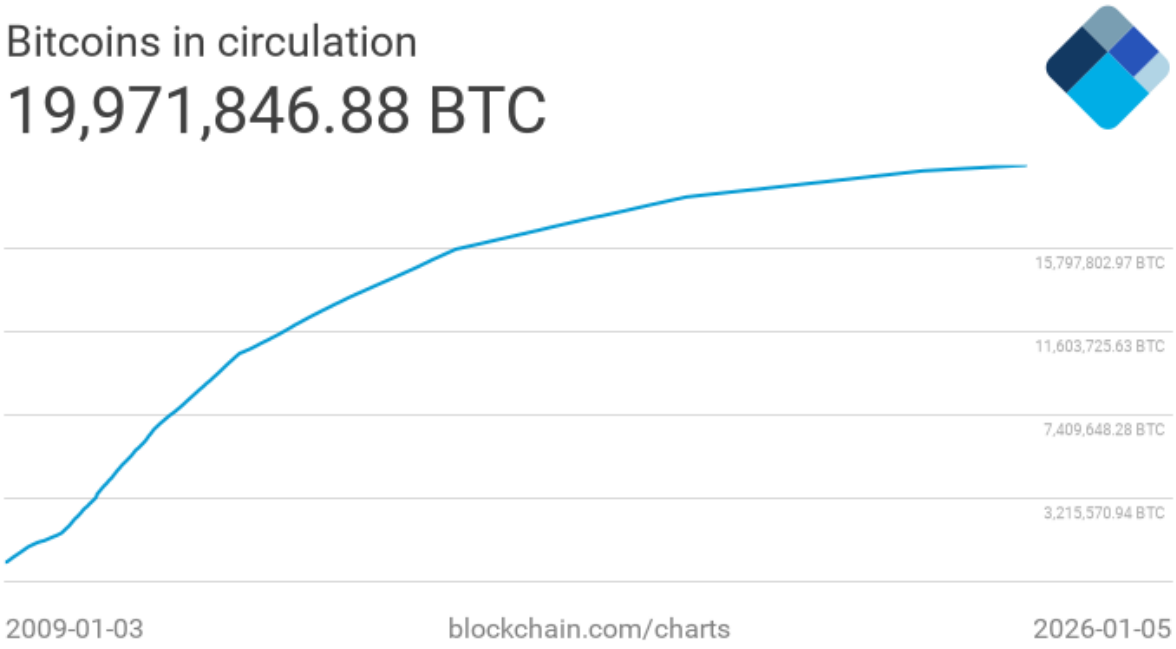
Introduction: The Bitcoin Mining Landscape

Bitcoin mining is the computational process that secures the bitcoin network and introduces new $BTC into circulation. It relies on a proof-of-work consensus mechanism in which miners compete to solve difficult **SHA-256** based cryptography puzzles. The reward for solving each puzzle is currently ~**3.125 $BTC** plus some transaction fees.

Block reward formula

$$Reward_t = \frac{50}{2^{\lfloor \frac{\text{Current Block}}{210000} \rfloor}}$$

Bitcoin's issuance follows a fixed, anti-inflationary tokenomics model with a maximum supply of **21M bitcoins**. Approximately **19.8M bitcoins** (Blockchain.com)are in circulating supply and it is estimated that bitcoin mining will cease by the year 2140 with rewards for miners decreasing over time.



All-time total $BTC circulating supply. Source: Blockchain.com.

This is because Bitcoin supply is governed by halving economics. This means, the reward for bitcoin mining decreases by half every **210K blocks**- approximately every 4 years (Investopedia, 2025). As a result, miners depend on higher ASIC efficiencies plus reduced electricity costs to keep their profit margins, making energy costs a central determinant of profitability.

The hardware used to mine bitcoin can range from: application specific integrated circuits (ASICs,specialized hardware optimized for SHA-256 hashing) to CPUs and GPUs. ASICs are the most widely used due to their higher efficiency and higher hash rate output. Hash rate in this context, refers to the rate at which an ASIC or bitcoin mining tool can solve the aforementioned cryptography problems-whose difficulty increases with time.

Mining Difficulty

$$Difficulty_t = Difficulty_{t-1} \times \frac{\text{Time for 2016 Blocks}}{\text{2 Weeks}}$$

Hash rate is a key performance indicator in determining profitability for Bitcoin miners and is measured in H/s(Hashes per second). Typical ASIC efficiency is also measured in Joules per Terahash(J/TH) (Bashari et al., 2025).

While ASIC efficiency continues to improve, mining hardware requires significant energy input and contributes to global e-waste as older machines become obsolete(Bashari et al., 2025). Notably, companies like

Mara Holdings have done much to ensure any excess energy from bitcoin mining can be rerouted to heat homes, as seen with their pilot project-heating around **11,000** homes in Finland.

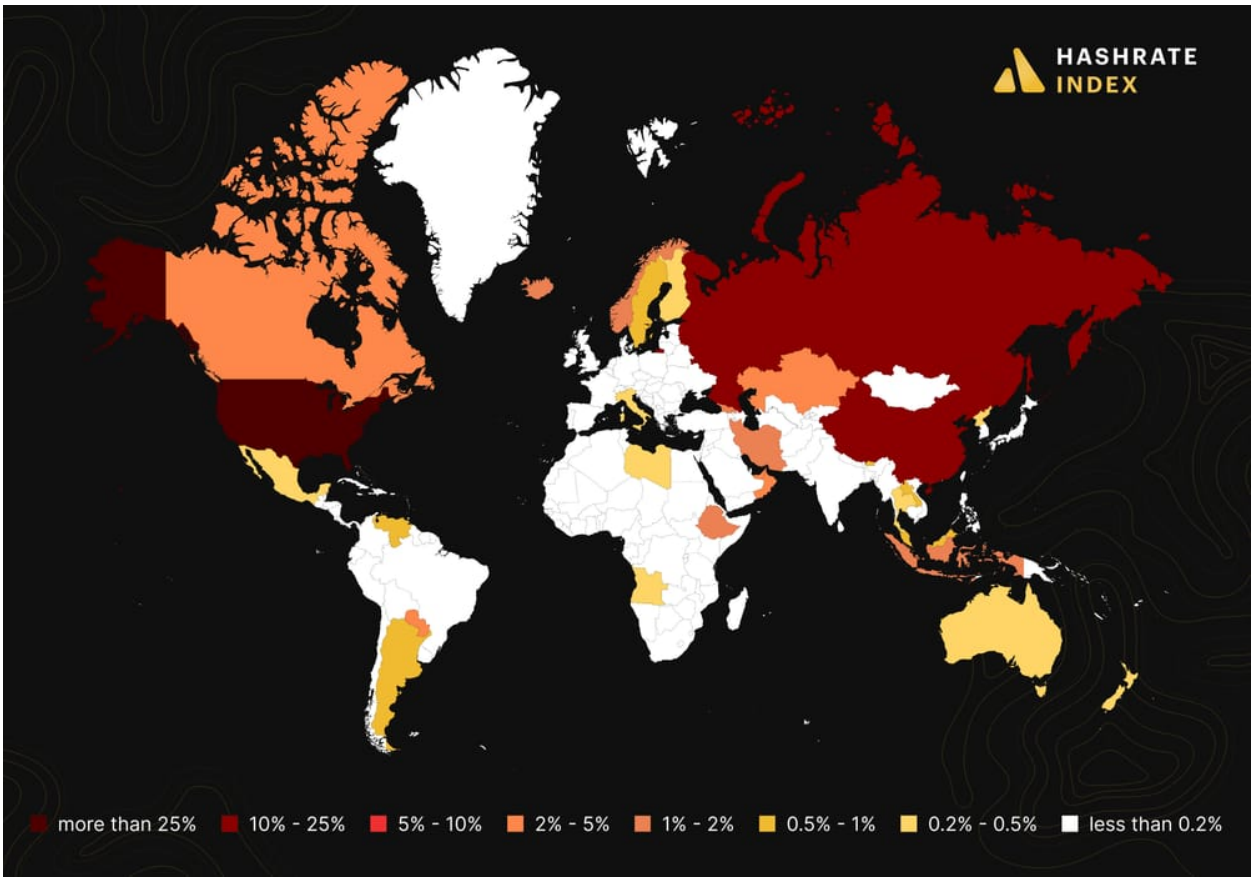
Miners face increasing regulatory pressure related to sustainability, grid impact & congestion and emissions. As block rewards decline and competition for hash rate intensifies, the industry is shifting toward low-cost, stable and scalable renewable energy sources as a strategic necessity alongside an eventual reliance on transaction fees.

The Energy Challenge in Bitcoin Mining

Mining bitcoin carries a range of operational expenses. These include:

- ASICs procurement and depreciation
- Cooling infrastructure and thermal management
- Labor and maintenance costs
- Housing and land costs
- Electricity costs which make up **60-80%** of OPEX depending on the miner.

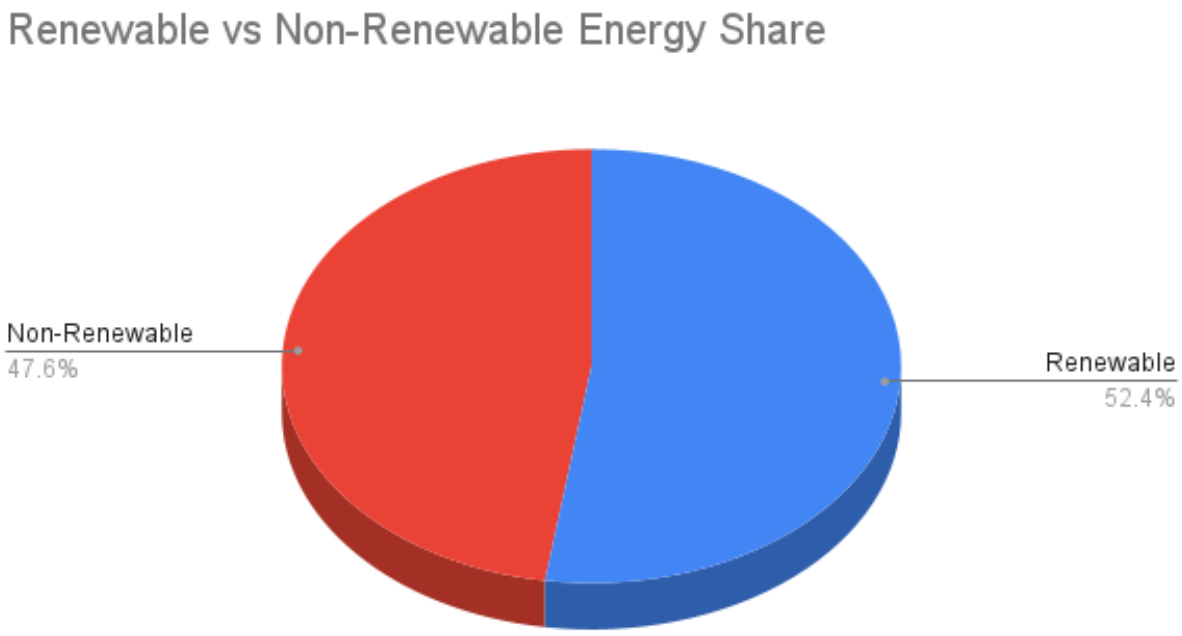
Electricity costs measured in $/kWh, range significantly by geo-location. Before 2022, most bitcoin mining operations were carried out in China(60% of the hash rate) (Bitbo.io)due to cheap coal-based electricity backed by high government subsidies as well as a Hydro-electric power mix. However, following regulatory crackdowns on energy-intensive activities and concerns over grid stability and emissions, bitcoin mining was outlawed. Today, miners operate in different countries like: Iran, Kazakhstan, some parts of China and Russia, some U.S.A states, parts of Africa and Latin America(hashrateindex.com, 2025).



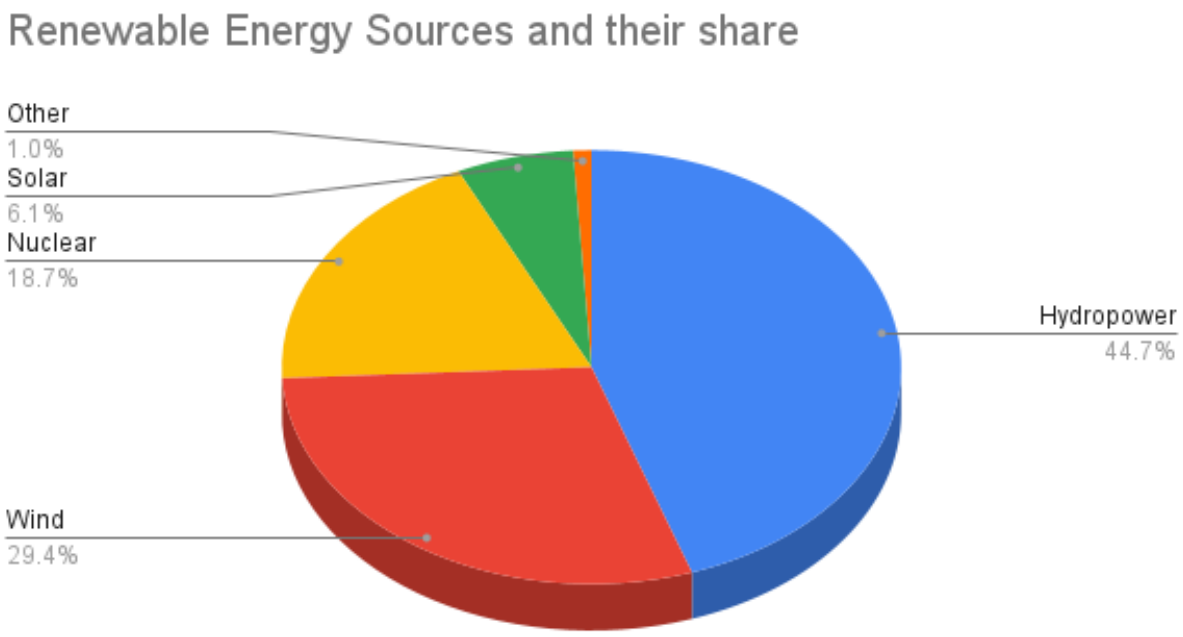
Global Hashrate Distribution(2025). Source: hashrateindex.com.

Energy is cheapest in regions like Iran and Kazakhstan which are coal-based and in some states like Texas and Kansas which offer incentives to bitcoin miners who relocate to them. Western countries like Ireland offer higher tariffs(electricity costs) for miners due to a higher residential usage/ demand for electricity(solartechonline, 2025) (Bashari et al., 2025).

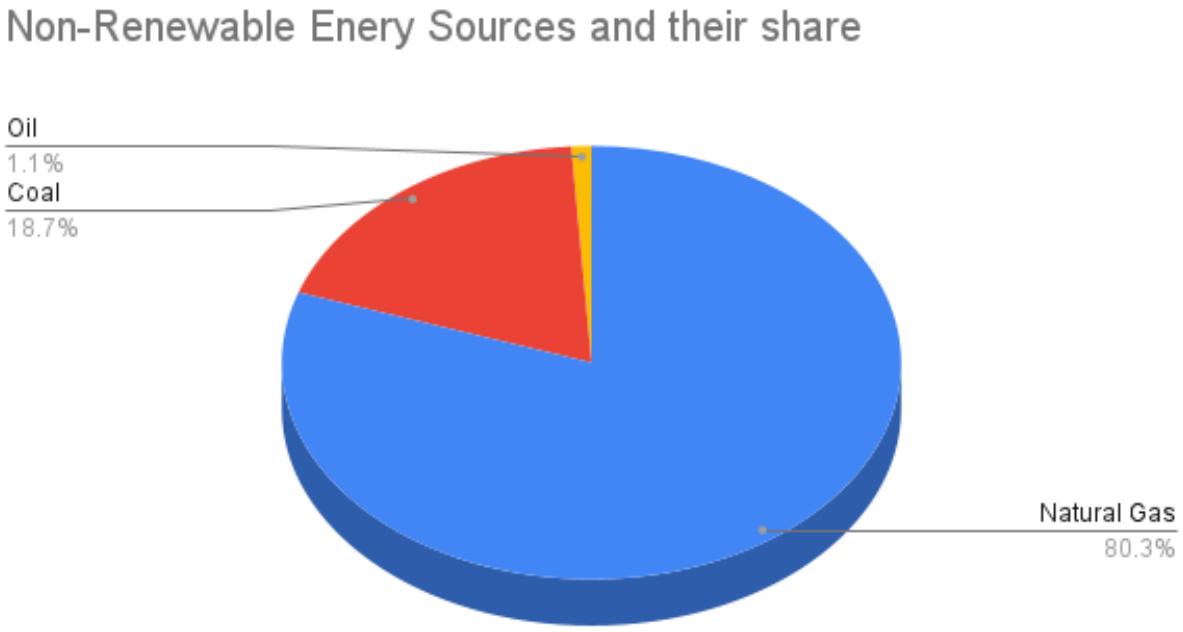
A Cambridge Digital mining report also stated that miners' electricity mix is predominantly sustainable **(52.4%)**, with renewables accounting for **42.6%**. Hydropower constitutes the largest sustainable source **(23.4%)**, followed by wind **(15.4%)**, nuclear **(9.8%)**, solar **(3.2%)**, and other renewables **(0.5%)**. Fossil fuels make up **47.6%**, primarily natural gas **(38.2%)**, which is also the single largest energy source, followed by coal **(8.9%)** and oil **(0.5%)**(Cambridge Digital Mining Industry Report, 2025).



Renewable vs Non-Renewable Energy Share(April 2025). Source: Cambridge Digital Mining Industry Report. *Visualization modified by author*



Renewable Energy Share(April 2025). Source: Cambridge Digital Mining Industry Report. *Visualization modified by author*



Non-Renewable Energy Share(April 2025). Source: Cambridge Digital Mining Industry Report. *Visualization modified by author*

Grid-connected mining operations face several structural risks:

- Public and Environmental Scrutiny: High energy use often leads to media attention and negative public perception, which can influence regulatory action and hinder business expansion plans. This can be seen with President Trump's recent announcement on Truth Social to ensure data centers 'pay their fair share' of electricity costs due to grid-connected participation(X.com, 2025).
- Uncertain Regulatory exposure: In the US, the U.S Energy Information Administration recently moved to initiate a provisional survey of electricity consumption for U.S cryptocurrency mining companies, which was paused after a series of those companies sued it (earthjustice.org, 2024).
- Forced shutoffs during high-demand periods: Grid-connected operations often have to shut down operations during high demand periods, to allow for energy use by more critical infrastructure and stabilize prices which has an effect on expected profitability(finance.yahoo.com, 2026).

Off-grid mining, by contrast, offers energy sovereignty. Miners that generate their own electricity avoid grid congestion, price spikes, and curtailment mandates, while gaining greater control over operating costs and regulatory exposure.

As block rewards decline and competition intensifies, energy economics not hardware alone-becomes the primary determinant of mining viability. Solar energy, particularly in regions with exceptional irradiance such as Botswana presents a strategic opportunity to address energy demands and mandates for sustainable energy use.

END OF FREE EXECUTIVE PREVIEW

You have reached the end of the public preview. To read the complete research paper, financial models, and operational blueprints, unlock the **Full Edition (Pages 8–17)**:

- **Section 4:** Botswana Solar Resource Data, DNI Potential & IPP Project Pipeline
- **Section 5:** $1.3M Solar Microgrid LCOE Financial Model & Monte Carlo Simulation ($0.071–$0.091/kWh)
- **Section 6:** Solar + AI Data Center + Bitcoin Mining Load-Stacking Architecture ($1M–$4M/MW Revenue Lift)
- **Section 7:** Risk Mitigation Matrix & 4-Phase Infrastructure Roadmap

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